

Problem 29 (Kurschak Competition 2014 Problem 1). There are $n \geq 4$ people in a meeting. Some pairs of people know each other. Suppose each person knows at least one other person but no one knows all other people. Prove that we can find 4 people and arrange them in a circle such that all of them knows exactly one of their two neighbours.

Solution. We regard this as a problem in graph theory in the obvious way. Let A be a vertex with the largest degree. Since no vertex has degree $n - 1$, there exists B which is not adjacent to A . Choose any adjacent vertex C to B .

Since $\deg C \leq \deg A$, there must be a neighbour D of A which is not a neighbour of C (no matter C is a neighbour of A or not). We can form the circle (A, B, C, D) that meets the requirement, because $\{A, D\}, \{B, C\}$ are edges while $\{A, B\}, \{C, D\}$ are not. $\square$

Problem 30 (Dutch IMO TST 2016 Day 3 Problem 1). Let n be a positive integer. In a village, n boys and n girls are living.

For the yearly ball, n dancing couples need to be formed, each of which consists of one boy and one girl. Every girl submits a list, which consists of the name of the boy with whom she wants to dance the most, together with zero or more names of other boys with whom she wants to dance. It turns out that n dancing couples can be formed in such a way that every girl is paired with a boy who is on her list.

Show that it is possible to form n dancing couples in such a way that every girl is paired with a boy who is on her list, and at least one girl is paired with the boy with whom she wants to dance the most.

Solution. We first form an arbitrary pairing such that every girl is paired with a boy on her list. If a girl is already paired with the boy with whom she wants to dance the most, we are done. WLOG assume each girl k is paired with boy k , and girl k wants to dance with boy $f(k) \neq k$ the most.

Since there is a finite number of people, there must be indices $i < j$ such that $f^i(1) = f^j(1)$. Let $k = f^i(1)$ and choose the smallest index $d > 0$ such that $f^d(k) = k$ (note that $f^{j-i}(k) = k$). Instead of the pairings $(k, k), (f(k), f(k)), \dots, (f^{d-1}(k), f^{d-1}(k))$ (with the first entry representing a girl and the second entry representing a boy), we form the pairings

$$(k, f(k)), (f(k), f^2(k)), \dots, (f^{d-1}(k), k),$$

together with other unaffected pairs. After the change, each girl is still paired with a boy on her list, and girl k is paired with the boy with whom she wants to dance the most. $\square$

Problem 31 (All-Russian Olympiad 2012 Grade 9 Day 2 Problem 4). There are some bus routes in a city. Each bus route has at least four stops, and any two bus routes have exactly one stop in common. Prove that the stops can be partitioned into two groups such that each bus route includes stops from each group.

Solution. If there exists a stop which is common to all bus routes, we can put this stop in the first group, and all other stops in the second group. Then each route consists of 1 stop in the first group and at least 3 stops in the second group because each route has at least 4 stops.

Suppose there does not exist a stop which is common to all bus routes. Consider an arbitrary route R_1 . WLOG assume route R_2 shares the stop A with R_1 , and route R_3 shares the common stop $B \neq A$ with R_1 . Let C be the common stop of R_2 and R_3 . We put A, B, C in the first group, the stops in R_1, R_2, R_3 other than A, B, C in the second group, and all other stops in the first group. We claim that this satisfies the requirement.

For each R_i , for example R_1 , it consists of two stops A, B in the first group, and at least two stops in the second group. For any other route, since it shares at most 3 stops with R_1, R_2, R_3 , it has at least one stop not in R_1, R_2, R_3 , which must be in the first group. Also, it consists of at most one of A, B, C . WLOG assume it does not consist of A, B . Then it must share a common stop with R_1 different from A, B , which belongs to the second group. This finishes the checking. $\square$

Problem 32 (Kurschak Competition 2008 Problem 3). In a country, there are at least two cities. We can travel from some cities to some other cities by plane or by train (possibly both). Some routes may operate in a single direction only. For every pair of cities A and B , we can go from A to B , or from B to A (or both) by travelling on only one type of vehicles (possibly through some other cities). Prove that there exists a city such that any other city can be reached by travelling on only one type of vehicles. (It is possible to reach different cities using different types of vehicles.)

Solution. We regard this as a problem in graph theory in the obvious way, with the directed edges coloured red and blue if they represent travels by plane and train respectively. Suppose there are $n \geq 2$ vertices. We prove by induction that for any $k \leq n$ of these vertices, there exists a *good vertex*, i.e. a vertex that can go to any other vertex in this group of k vertices using monochromatic edges, possibly via some vertices outside this group.

The base case $k = 1$ holds trivially, and the case $k = 2$ is true by assumption. Suppose all smaller cases hold. Consider any k vertices $V_1, V_2, \dots, V_k$. By the inductive hypothesis, WLOG assume V_1 is a good vertex among $V_1, V_2, \dots, V_{k-1}$. Suppose V_1 can reach $V_2, V_3, \dots, V_m$ using red edges, and reach $V_{m+1}, \dots, V_{k-1}$ using blue edges. If V_1 can reach V_k using monochromatic edges, we are done. WLOG we may assume V_k can reach V_1 using red edges.

Consider the group $V_{m+1}, \dots, V_k$ of $k - m \leq k - 1$ vertices. By the inductive hypothesis, there exists a good vertex in this group.

- If V_k is a good vertex in this group, then V_k can reach $V_{m+1}, \dots, V_{k-1}$, and can also reach V_1 and then $V_2, \dots, V_m$ using red edges. So V_k is a good vertex of the group $V_1, V_2, \dots, V_k$.
- If V_k is not a good vertex in this group, WLOG assume V_{m+1} is a good vertex in this group. If it can reach V_k using red edges, then V_{m+1} can reach $V_{m+2}, \dots, V_{k-1}$, and can also reach V_k and then V_1 and then $V_2, \dots, V_m$ using red edges. So V_{m+1} is a good vertex of the group $V_1, V_2, \dots, V_k$. If V_{m+1} can reach V_k using blue edges, then V_1 can reach $V_2, \dots, V_{k-1}$, and can also reach V_m and then V_k using blue edges. So V_1 is a good vertex of the group $V_1, V_2, \dots, V_k$.

We have found a good vertex of $V_1, V_2, \dots, V_k$ in any case. This completes the inductive step. By induction, the case $k = n$ also holds, as desired. $\square$